Arduino Simulator Coverage Summary

Workspace source coverage summary generated on March 27, 2026 for `/Users/fred/Documents/arduino_simulator`. The run used instrumented Rust tests across the full workspace. Percentages below are for project source files under `*/src/*.rs`, not test files.

SOURCE FILES
36 Project files included in the rollup

WORKSPACE TESTS
171 All passing in the instrumented run

COVERAGE COMMAND
`cargo test --workspace` With `-Cinstrument-coverage`

ARTIFACTS
`coverage/` PDF, HTML, and TSV included

Topline

Overall source line coverage is 60.53% with `8,775 / 14,496` lines covered. Function coverage is `66.21%` and region coverage is `60.07%`.

METRIC	COVERED	TOTAL	PERCENT
Lines	8,775	14,496	60.53%
Functions	817	1,234	66.21%
Regions	12,937	21,537	60.07%

What Stands Out

- **Best-covered crates:** `rust_board` at 94.28% line coverage, `rust_mcu` at 85.06%, and `rust_project` at 80.20%.
- **Middle of the pack:** `rust_cpu` is at 69.74%, which is much healthier than before but still has one very large hotspot in `cpu.rs`.
- **Main drag on the workspace total:** `rust_gui` and `rust_runtime` have the most uncovered source lines in absolute terms.
- **Most concerning single file:** `rust_project/src/error.rs` is at 0.00%, which usually means an entire error/reporting surface is not exercised in tests yet.

Coverage By Crate

CRATE	LINE %	COVERED / TOTAL LINES	FUNCTION %	COVERED / TOTAL FUNCTIONS	REGION %
rust_board	94.28%	1,566 / 1,661	94.08%	159 / 169	93.49%
rust_mcu	85.06%	1,002 / 1,178	82.05%	64 / 78	83.66%
rust_project	80.20%	1,207 / 1,505	86.30%	126 / 146	78.87%
rust_cpu	69.74%	1,302 / 1,867	98.44%	63 / 64	68.99%
rust_behavior	57.14%	412 / 721	75.00%	36 / 48	52.27%
rust_runtime	45.60%	528 / 1,158	48.72%	57 / 117	46.91%
rust_gui	43.05%	2,758 / 6,406	50.98%	312 / 612	43.75%

Biggest Hotspots By Missed Lines

FILE	COVERAGE	MISSED LINES	COVERED / TOTAL
rust_gui/src/app.rs	45.52%	1,460	1,220 / 2,680
rust_gui/src/board_editor.rs	39.97%	1,218	811 / 2,029
rust_cpu/src/cpu.rs	67.21%	565	1,158 / 1,723
rust_runtime/src/tui.rs	26.78%	462	169 / 631
rust_gui/src/simulation.rs	21.65%	445	123 / 568
rust_gui/src/pcb_view.rs	53.49%	440	506 / 946
rust_behavior/src/lib.rs	56.30%	298	384 / 682
rust_runtime/src/cli.rs	68.99%	89	198 / 287
rust_gui/src/arduino.rs	53.55%	85	98 / 183
rust_mcu/src/atmega2560.rs	84.35%	82	442 / 524

Lowest-Coverage Files

FILE	COVERAGE	MISSED LINES
rust_project/src/error.rs	0.00%	53
rust_gui/src/simulation.rs	21.65%	445
rust_runtime/src/tui.rs	26.78%	462
rust_gui/src/board_editor.rs	39.97%	1,218
rust_runtime/src/firmware.rs	42.59%	62

FILE	COVERAGE	MISSED LINES
rust_gui/src/app.rs	45.52%	1,460
rust_gui/src/pcb_view.rs	53.49%	440
rust_gui/src/arduino.rs	53.55%	85
rust_behavior/src/lib.rs	56.30%	298
rust_project/src/shared.rs	65.26%	33

Strongest-Coverage Files

FILE	COVERAGE	MISSED LINES
rust_board/src/layout.rs	100.00%	0
rust_cpu/src/bus.rs	100.00%	0
rust_cpu/src/config.rs	100.00%	0
rust_cpu/src/error.rs	100.00%	0
rust_cpu/src/lib.rs	100.00%	0
rust_mcu/src/lib.rs	100.00%	0
rust_runtime/src/main.rs	100.00%	0
rust_board/src/kicad_pcb.rs	96.39%	10
rust_board/src/models.rs	96.27%	15
rust_board/src/kicad_layout.rs	94.81%	32

Recommended Next Targets

- **GUI depth:** If the goal is the fastest improvement to workspace line coverage, the best return is in `rust_gui/src/app.rs` and `rust_gui/src/board_editor.rs` because they combine low percentages with very large line counts.
- **Runtime confidence:** `rust_runtime/src/tui.rs` and `rust_runtime/src/firmware.rs` are under-tested relative to the rest of the runtime package.
- **Error handling:** `rust_project/src/error.rs` should be easy to lift from 0% with a compact formatting/variant test file.
- **CPU depth:** `rust_cpu/src/cpu.rs` is no longer the main drag on the workspace total, but it is still one of the largest uncovered files in absolute missed lines.

Per-File Appendix

FILE	LINE %	COVERED / TOTAL LINES	MISSED
rust_behavior/src/builtins.rs	71.79%	28 / 39	11
rust_behavior/src/lib.rs	56.30%	384 / 682	298
rust_board/src/dsl.rs	73.33%	33 / 45	12
rust_board/src/kicad_layout.rs	94.81%	584 / 616	32
rust_board/src/kicad_pcb.rs	96.39%	267 / 277	10
rust_board/src/lang.rs	90.29%	158 / 175	17
rust_board/src/layout.rs	100.00%	25 / 25	0
rust_board/src/models.rs	96.27%	387 / 402	15
rust_board/src/sexpr.rs	92.56%	112 / 121	9
rust_cpu/src/bus.rs	100.00%	49 / 49	0
rust_cpu/src/config.rs	100.00%	54 / 54	0
rust_cpu/src/cpu.rs	67.21%	1,158 / 1,723	565
rust_cpu/src/error.rs	100.00%	21 / 21	0
rust_cpu/src/lib.rs	100.00%	20 / 20	0
rust_gui/src/app.rs	45.52%	1,220 / 2,680	1,460
rust_gui/src/arduino.rs	53.55%	98 / 183	85
rust_gui/src/board_editor.rs	39.97%	811 / 2,029	1,218
rust_gui/src/pcb_view.rs	53.49%	506 / 946	440
rust_gui/src/simulation.rs	21.65%	123 / 568	445
rust_mcu/src/atmega2560.rs	84.35%	442 / 524	82
rust_mcu/src/atmega328p.rs	82.88%	363 / 438	75
rust_mcu/src/common.rs	86.13%	118 / 137	19
rust_mcu/src/lib.rs	100.00%	79 / 79	0
rust_project/src/assembly_bundle.rs	87.93%	204 / 232	28
rust_project/src/behavior_definition.rs	81.36%	96 / 118	22
rust_project/src/definitions.rs	86.03%	388 / 451	63
rust_project/src/document.rs	70.21%	33 / 47	14
rust_project/src/error.rs	0.00%	0 / 53	53
rust_project/src/paths.rs	79.17%	285 / 360	75

FILE	LINE %	COVERED / TOTAL LINES	MISSED
`rust_project/src/shared.rs`	65.26%	62 / 95	33
`rust_project/src/simulation_project.rs`	93.29%	139 / 149	10
`rust_runtime/src/cli.rs`	68.99%	198 / 287	89
`rust_runtime/src/firmware.rs`	42.59%	46 / 108	62
`rust_runtime/src/main.rs`	100.00%	6 / 6	0
`rust_runtime/src/runtime.rs`	86.51%	109 / 126	17
`rust_runtime/src/tui.rs`	26.78%	169 / 631	462

Method: instrumented workspace test run with `RUSTFLAGS='-Cinstrument-coverage'`, merged via `xcrun llvm-profdata`, summarized from `xcrun llvm-cov export -summary-only`.